Supplementary Information

Decision rules rather than representations drive external failures of diabetic retinopathy grading models

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Supplementary Table S1. Reporting checklist (TRIPOD+AI and CLAIM)

The study develops and externally evaluates prediction models (TRIPOD+AI) built from medical images (CLAIM). The table maps each reporting domain to the place where it is addressed. The journal's official checklist forms are to be completed by the authors at submission from this mapping.

Domain	Where and how it is reported
Title and abstract	Title names the task (DR grading) and the finding; abstract states data, models, external sets, decision rules and main results.
Background and objectives	Introduction: the gap (external failures attributed to representations without matched decision rules) and the pre-registered questions.
Data sources	Methods, *Data*; Table 1. Public corpora: IDRiD, Messidor-2 (adjudicated grades), EyePACS (pretraining only), APTOS 2019, DDR.
Participants and eligibility	Methods, *Data*: every gradable image with a DR grade 0–4; images with DDR grade 5 (ungradable) excluded. No patient-level demographics are released by the corpora; this is stated as a limitation.
Data preparation	Methods, *Models and training*: retinal-disc crop, cache at long side 560 pixels, resizing, augmentation.
Outcome (reference standard)	International 5-grade DR scale; DME risk grade (IDRiD); grading procedure per corpus in Table 1.
Sample size	Fixed by the public corpora; uncertainty reported by bootstrap intervals throughout.
Missing data	Messidor-2 has no three-grade DME label; the DME loss is masked for those images (Methods).
Data partitions	Methods, *Cross-validation and model selection*: frozen 5-fold split of the development pool; external sets never used for training, selection or threshold fitting except the explicitly diagnostic R4/R4m. Content-overlap audit: Supplementary Note 7.
Model specification	Methods and Supplementary Table S2 (every configuration, one factor changed per run).
Model output and decision rule	Methods, *Decision rules*; Supplementary Note 1.
Performance measures	QWK, Cohen's κ, accuracy, per-grade recall, macro-averaged recall, referable-DR sensitivity and specificity, AUROC, Brier score, ECE (Methods, *Statistics*).
Uncertainty	Paired bootstrap over images (2,000 resamples) for every comparison.
Class imbalance	DR-grade-balanced sampler (Methods); per-grade recall reported for every grade.
Fairness	No subgroup attributes are released by the corpora; between-population behaviour is studied through four external populations in two directions.
Pre-registration	Outcomes O1–O9 and the interventions were registered in the public repository before the corresponding runs (Methods, *Pre-registration*); falsified outcomes are reported.
Code and data availability	Data availability and Code availability statements.
Ethics	Ethics statement (public, de-identified data).
Use of AI assistance	Methods, final paragraph.
Limitations	Discussion.

Supplementary Table S2. Configurations

Table S2: Configurations. Each configuration changes one factor relative to its control. Dev QWK: development out-of-fold DR QWK at matched calibration (fold-wise cross-fitted cut-points); folds: development folds trained.

Model	Control	Backbone	DR head	Pixels	Pretraining	Epochs	Factor
E08	–	DenseNet121	cumulative	448	ImageNet + EyePACS (5 ep)	40	reference
E09	E08	DenseNet121	cumulative	224	ImageNet + EyePACS	40	resolution
E10	E08	DenseNet121	cumulative	640 (560 effective)	ImageNet + EyePACS	40	resolution
E17NAT	E10	DenseNet121	cumulative	640	ImageNet + EyePACS	40	native-res
E11FULL	E10	EfficientNet-B3	cumulative	448	ImageNet + EyePACS	30	backbone
E14MAC	E08	DenseNet121	cumulative	448 + macula crop	ImageNet + EyePACS	40	fovea-cent
E15LPFT	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40 (LP 5)	linear pro
E20CORAL	E08	DenseNet121	CORAL	448	ImageNet + EyePACS	40	ordinal he
E22CORN	E08	DenseNet121	CORN	448	ImageNet + EyePACS	40	ordinal he
E21CNXA	E08	ConvNeXt-tiny	cumulative	448	ImageNet + EyePACS	40	backbone
I24FULL	E09	RETFound ViT-L/16	cumulative	224	RETFound (colour fundus)	30 (LP 5)	foundatio
S1E08–S4E08	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40	seed only
R1APTOS	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40	trained or

Supplementary Table S3. Pre-registered interventions

Table S3: Pre-registered interventions, shipped and matched calibration. Difference (intervention minus control) in development DR QWK with 95% paired-bootstrap intervals (DME QWK for the macula crop; APTOS DR QWK for the ensemble).

Intervention	Control	Folds	Matched difference [95% CI]	Verdict
EfficientNet-B3 backbone	E10	5	+0.0207 [+0.0105, +0.0320]	confirmed
Macula-centred crop (DME QWK)	E08	5	−0.0237 [−0.0566, +0.0094]	falsified
Linear probe then fine-tune	E08	5	+0.0008 [−0.0105, +0.0126]	falsified
Native 640-px resolution	E10	5	−0.0086 [−0.0199, +0.0025]	falsified
CORAL ordinal head	E08	5	−0.0076 [−0.0184, +0.0032]	falsified
CORN ordinal head	E08	5	−0.0082 [−0.0207, +0.0036]	falsified
ConvNeXt-tiny backbone	E08	2	−0.0362 [−0.0593, −0.0153]	falsified, significantly worse
RETFound ViT-L/16 (fine-tuned)	E09	2	−0.0399 [−0.0669, −0.0124]	falsified, significantly worse
Six-model ensemble (APTOS, external)	E10	5	+0.0025 [−0.0016, +0.0066]	falsified

Supplementary Note 1. Decision rules

Every model emits, for DR, four logits whose sigmoids estimate $P(y > k)$, $k = 0, \ldots, 3$ (cumulative head and CORAL), or conditional probabilities $P(y > k \mid y > k-1)$ that are chained by cumulative product (CORN). From these the expected grade $s = \sum_k P(y > k)$ is computed. The five rules are:

R1 (shipped). Grade $= \#\{k : P(y > k) > 0.5\}$ after enforcing monotonicity by a running minimum. This is what the trained model outputs if deployed unchanged.

R2 (development QWK cut-points). Four ordered cut-points on s chosen to maximise QWK on the development out-of-fold predictions, by coordinate ascent over a grid of 60 candidate values per cut-point (8 rounds), then applied unchanged externally.

R3 (R2 with a referral constraint). R2 with the grade-1/2 cut-point replaced by the largest value that keeps development referable-DR sensitivity at or above 0.90, the neighbouring cut-points pushed to preserve order.

R4 and R4m (external, diagnostic). The same search on the external labels, with the external set split at random into two halves (seed 0); cut-points fitted on one half are applied to the other and vice versa, so no image is graded by cut-points fitted on its own label. R4 maximises QWK, R4m macro-averaged recall. The grid has 80 candidate values.

R4/R4m use external labels and are never reported as performance: they bound what a local recalibration of the decision rule alone, without retraining, can recover. Changing the development grid from 60 to 80 values changes E08's matched development QWK by 0.0033; 60 is used for all development quantities because it reproduces the archived matched comparisons exactly.

Supplementary Note 2. The DME grade cannot be derived from published lesion masks

IDRiD defines its DME risk grade by the distance from the fovea to the nearest hard exudate: grade 0, no exudate; grade 1, exudates more than one disc diameter from the fovea; grade 2, within one disc diameter. Because the definition is geometric, a grade derived from the pixel-level hard-exudate masks that IDRiD releases for 81 images, together with its fovea and optic-disc annotations, should reproduce the expert grade if the grade is what its definition says. The protocol was pre-registered (`docs/PARTA_preregistration.md`).

Of the 81 segmentation images, 54 could be linked to their grading-set copies by pixel correlation (minimum correlation 0.9999). The disc diameter was measured from the optic-disc mask in three ways (major axis, minor axis, equivalent-area diameter); all three give the same grades. The derived grade matched the expert grade in 96.3% of images, against 92.6% for a constant predictor, because almost every segmentation image is grade 2. On the 4 images where the test is informative (expert grade 0 or 1), the derivation failed twice: an image graded 1 whose nearest exudate lies 53 pixels from the fovea (disc diameter 612 pixels), and an image graded 0 with a single exudate 87 pixels from the fovea (disc diameter 657 pixels). In both, a single small exudate inside the threshold is either not counted by the grader or not visible at grading resolution. The pre-registered gate (exact agreement above the constant predictor on the informative images) therefore failed, and no derived DME labels were used anywhere.

Supplementary Note 3. A per-grade verdict that depends on the cut-point procedure

The development record that motivated the matched-calibration rule included a source control (E12): two runs identical except for whether Messidor-2 images came from the default or the

native-resolution files, compared on the 953 development images both could use. Moderate recall, native minus default, under four ways of setting both arms' cut-points:

Cut-points for both arms	Moderate recall difference (points) [95% CI]
Shipped (0.5)	−6.9 [−12.1, −2.1]
Fitted on all rows (in-sample)	−10.8 [−16.2, −5.4]
Random two-fold cross-fit, grid 80	−6.6 [−12.0, −1.1]
Fold-wise cross-fit, grid 60 (development standard)	+0.4 [−5.7, +6.1]

The originally archived value, which the project had cited as the case in which a difference *survived* matched calibration, is reproduced by the random-half procedure; under the fold-wise procedure used for every other development comparison it dissolves. Aggregate QWK is indistinguishable under every procedure. Two consequences are drawn in the main text: a matched-calibration claim must name its fitting procedure, and per-grade claims should be reported under at least two. The two-directional property of the rule (that it can confirm as well as dissolve a difference) rests instead on EfficientNet-B3 versus the 560-pixel DenseNet121, which survives matched calibration on the development pool (+0.0202 shipped, +0.0207 [+0.0105, +0.0320] matched), and on per-grade external differences that stay significant at every rule (Table 3 of the main text).

Supplementary Note 4. Correction of an archived ordinal-head comparison

Regenerating every development number from the archived per-image logits with independent code reproduced each archived matched QWK exactly except one. The archived comparison of the CORN head against the reference reported −0.0479 [−0.0656, −0.0325], a significant deficit, with matched QWK 0.8168 for CORN. The archived script had decoded CORN's conditional logits as if they were cumulative; decoding them by cumulative product, as the training code does, gives matched QWK 0.8564 and a difference of −0.0082 [−0.0207, +0.0036], indistinguishable from the reference. The pre-registered verdict (no improvement) is unchanged; the claim that CORN is worse, and the mechanism once offered for it, are withdrawn. The script was corrected to read each run's head from its results file (repository `ISSUES.md`, §30).

Supplementary Note 5. Seed variation (O1–O3)

The reference configuration was trained with **??seeds.n??** different seeds on identical data, split and recipe. Matched development QWK ranged **??seeds.dev.lo??**–**??seeds.dev.hi??** (s.d. **??seeds.dev.sd??**); the yardstick for a difference beyond seed noise between two configurations is therefore $2\sqrt{2}$ s.d. = **??seeds.yardstick??** (grey band in Fig. 6). On APTOS under the shipped rule, QWK ranged **??seeds.aptos.qwklo??**–**??seeds.aptos.qwkhi??** (s.d. **??seeds.aptos.sd??**). Of **??seeds.aptos.ntests??** per-grade recall differences between pairs of seeds, **??seeds.aptos.sigR1??** had a bootstrap interval excluding zero under R1 (largest absolute difference **??seeds.aptos.maxR1??** points) and **??seeds.aptos.sigR4??** under R4: the pre-registered prediction O2 that seeds alone produce more than five "significant" per-grade differences **??seeds.o2??**. Mild recall under the shipped rule was **??seeds.aptos.mildR1.lo??**–**??seeds.aptos.mildR1.hi??**% and under R4m **??seeds.aptos.mildR4m.lo??**–**??seeds.aptos.mildR4m.hi??**% in every seed (O3 **??seeds.o3??**).

Table S4: Seed variation of the reference configuration (O1–O3). Five independent training seeds of E08 (identical data, split and recipe).

[seeds.json: analysis output not yet available.]

Supplementary Note 6. Reverse direction (O4–O6)

To test whether the mild-grade collapse is specific to transfer from the development pool, the reference recipe was trained on APTOS 2019 (its own frozen 5-fold split, near-duplicates grouped) and evaluated on Messidor-2 and IDRiD, the direction in which a published ordinal-regression model was reported to lose the mild grade. Matched out-of-fold QWK on APTOS was **??rev.dev.qwk??**. On Messidor-2 under the shipped rule mild recall was **??rev.m2.R1.mild??**% (O4 **??rev.o4??**); external cut-points (R4m) raised it to **??rev.m2.R4m.mild??**% without retraining (O5 **??rev.o5??**). The mean predicted minus true grade under the shipped rule was **??rev.m2.R1.mean??** (O6, under-grading in the reverse direction, **??rev.o6??**).

Table S5: Reverse direction (O4–O6). The reference recipe trained on APTOS 2019 and evaluated on Messidor-2 and IDRiD.

[reverse.json: analysis output not yet available.]

Supplementary Note 7. Content-overlap audit

Public fundus corpora are assembled from overlapping sources, so a near-duplicate of an external image in the training data would inflate external performance. Every image of the development pool, APTOS, DDR and EyePACS was reduced to a 256-bit difference hash after the same retinal crop used for training, and all cross-corpus pairs within **??ov.bits??** bits were counted (in this project's data, distinct images lie at least 28 bits apart and re-encodings of one image within 8). The development pool and the frozen split were built before this audit; development near-duplicate pairs that straddle two folds number **??ov.crossfold??**.

Table S6: Content-overlap audit. Pairs of images across corpora whose 256-bit difference hashes lie within the stated Hamming distance.

$$[overlap_audit.json : analysisoutputnotyetavailable.]$$

Supplementary Tables S4–S5. Full decision-rule matrices

Table S7: Full decision-rule matrix, APTOS. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.897	73.5	97.0	5.9	64.2	64.2	52.2	99.5	84.3
	R2	0.883	68.2	98.3	8.9	39.0	56.0	65.4	99.1	85.1
	R3	0.882	67.9	98.3	4.6	39.2	56.0	65.4	99.3	84.2
	R4	0.904	78.2	98.8	18.1	77.9	49.7	47.8	96.6	87.0
	R4m	0.886	75.2	96.6	70.8	49.0	47.2	56.3	80.4	95.4
E09	R1	0.879	69.3	88.5	4.3	66.8	62.7	45.8	99.3	84.1
	R2	0.863	62.0	88.5	22.7	30.3	62.2	56.6	96.3	87.4
	R3	0.847	61.0	88.5	1.4	34.3	62.2	56.6	99.7	79.7
	R4	0.898	76.7	98.6	10.0	75.5	48.7	48.5	98.6	85.9
	R4m	0.886	75.6	97.8	61.6	53.6	50.3	48.5	81.1	94.7
E10	R1	0.887	70.8	94.4	4.3	59.9	72.5	45.8	99.0	84.2
	R2	0.885	70.1	98.1	7.0	50.3	42.0	63.1	98.8	84.7
	R3	0.884	69.8	98.1	3.2	50.8	42.0	63.1	99.2	84.0
	R4	0.898	76.8	98.0	13.5	75.5	50.3	49.2	97.2	85.9
	R4m	0.876	73.0	97.8	77.8	36.6	50.8	52.5	72.6	97.0
E11FULL	R1	0.903	74.9	98.1	4.9	65.6	63.2	60.3	99.3	84.3
	R2	0.900	72.9	98.9	7.6	54.5	63.2	64.1	99.0	84.9
	R3	0.900	72.8	98.9	7.0	54.6	63.2	64.1	99.1	84.8
	R4	0.907	79.1	98.6	16.8	81.8	39.4	54.6	96.5	86.6
	R4m	0.890	73.4	97.6	75.1	37.3	51.3	59.7	75.3	96.4
E14MAC	R1	0.880	68.7	95.5	5.7	53.9	72.5	31.9	98.7	84.3
	R2	0.879	67.2	98.6	5.7	39.6	72.0	42.7	98.3	84.7
	R3	0.877	67.1	98.6	1.4	40.9	72.0	42.7	99.2	83.6
	R4	0.889	74.6	98.9	10.3	69.0	52.8	40.3	97.6	85.7
	R4m	0.857	71.5	97.4	75.7	35.7	44.0	46.8	71.2	96.6
E15LPFT	R1	0.874	65.3	89.6	5.1	45.1	70.5	57.6	99.2	84.2
	R2	0.880	69.2	97.2	10.3	43.5	57.0	66.1	98.3	85.3
	R3	0.878	68.9	97.2	5.1	44.5	57.0	66.1	99.1	84.2
	R4	0.898	77.3	98.6	11.9	77.5	44.6	49.2	97.3	86.0
	R4m	0.878	75.5	97.7	70.3	49.3	34.2	61.4	78.5	96.0
E17NAT	R1	0.891	71.1	96.1	5.9	56.8	72.0	47.8	99.2	84.6
	R2	0.872	65.5	97.3	9.5	32.2	48.2	64.7	99.0	85.0
	R3	0.870	64.9	97.3	3.2	32.6	48.2	64.7	99.4	83.7
	R4	0.894	76.5	99.1	16.2	72.0	48.7	47.8	96.3	86.9
	R4m	0.874	73.6	97.7	72.7	41.7	50.3	51.2	74.0	96.3

Table S8: Full decision-rule matrix, DDR. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

$$[phase0_ddr.json : analysisoutputnotyetavailable.]$$